SELF-PLAY REINFORCEMENT LEARNING · IMPERFECT-INFORMATION GAMES

AlphaHoldem-inspired self-play RL for Heads-Up No-Limit Hold'em

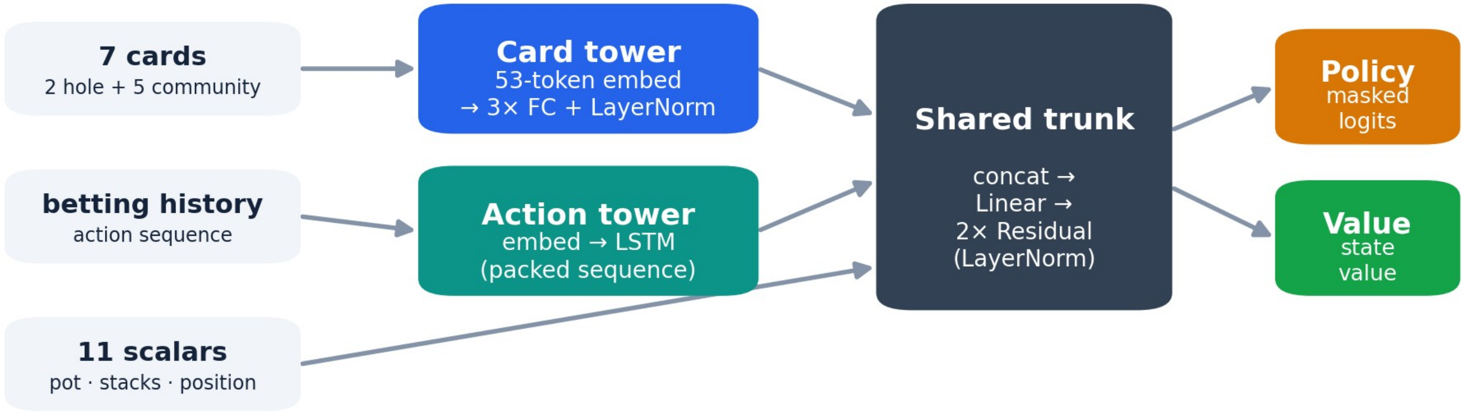
From-scratch PyTorch self-play agent · evaluation harness with behavior gates · exact-exploitability study on Leduc

Sanil Desai · PyTorch · OpenSpiel · single GPU

ARCHITECTURE

Pseudo-siamese two-tower actor–critic, ~1.5M parameters.

PPO + GAE · K-best self-play league · FCPA (4) / FCHPA (5) abstraction · OpenSpiel · single GPU



search-free: one masked forward pass per decision · trained with PPO + GAE against a K-best self-play league

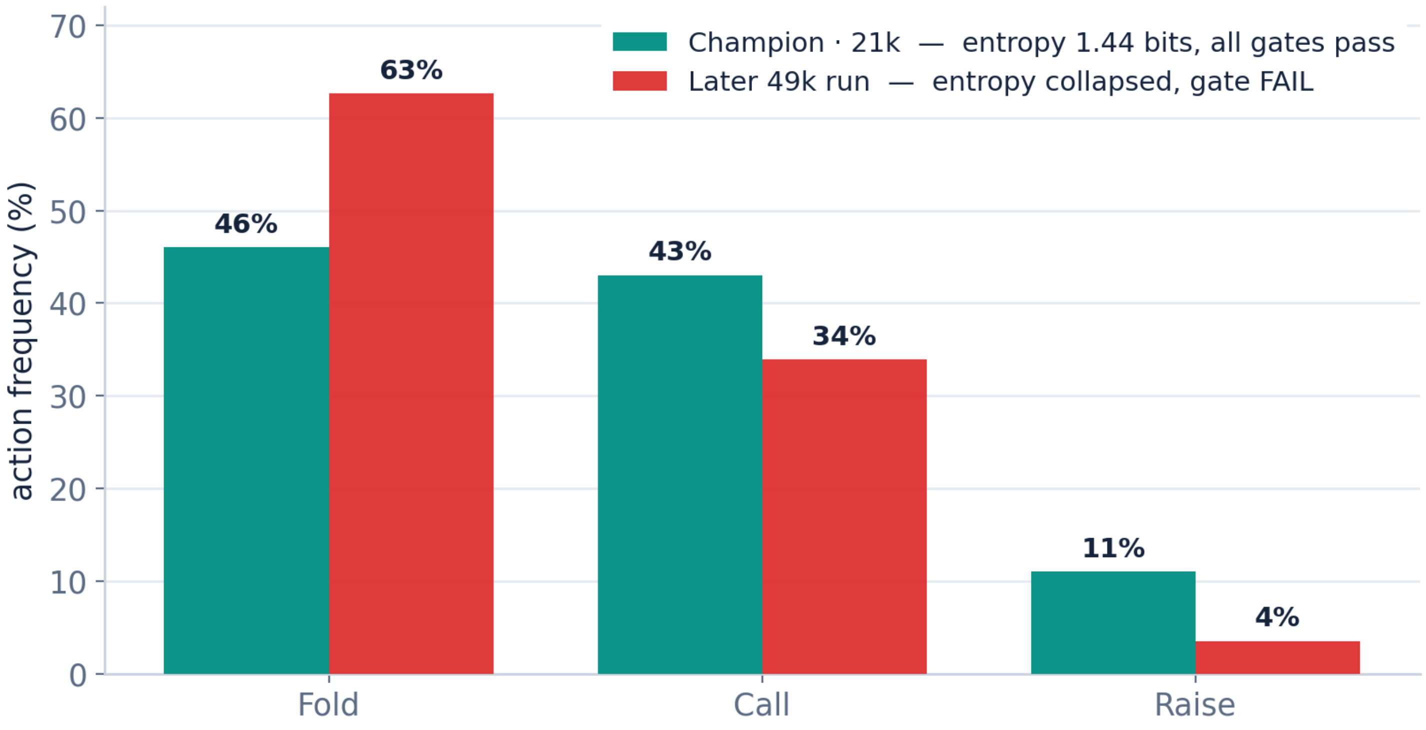
SCOPE

Inspired, not reproduced.

Axis	AlphaHoldem (paper)	This repo
Card / action input	multi-channel tensors → ConvNets	53-token embedding → MLP / LSTM
Loss	Trinal-Clip PPO	clipped PPO + Huber value
Self-play	top-K by Elo	top-K rolling + rating-temperature sampling
Game scope	abstraction-free full game	FCPA (4) / FCHPA (5)
Inference	search-free, single forward pass	**search-free, single forward pass ✓**
Parameters	~8.6M	~1.5M

Behavior gates gate promotion — and caught a policy collapse.

Envelopes on fold frequency, aggression, entropy, and bet-size mix. A later 49k run scored well on win-rate but collapsed to fold-heavy play; the gates blocked it.



BASELINE

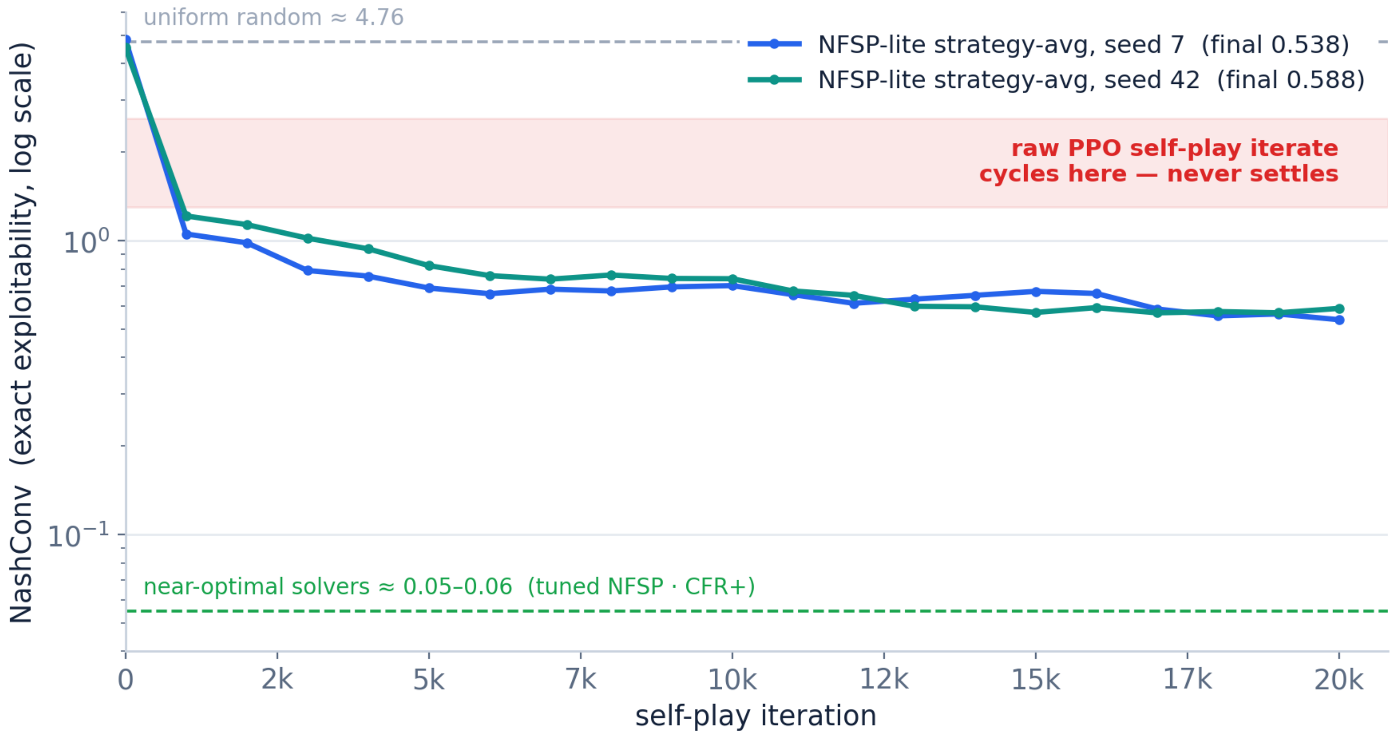
“+897 bb/100 vs MCCFR-ES” is vs a near-random control.

+970	+898	+934
vs 0-iteration uniform (random) · bb/100	vs 5k-iteration “solver” · bb/100	vs 10k-iteration “solver” · bb/100

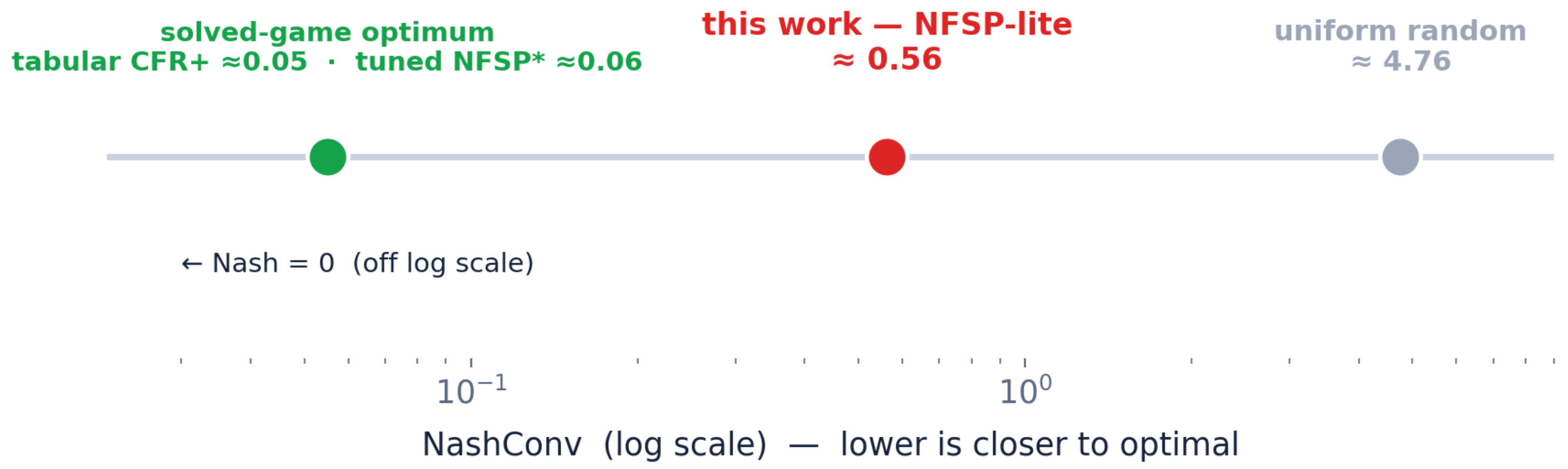
The uniform (0-iteration) control scores highest — adding “solver” iterations makes the opponent easier, not harder. The MCCFR-ES baseline is near-random; the figure is a sanity check, not strength, and is never placed next to AlphaHoldem’s +11 mbb/h.

Leduc (exact NashConv): the iterate cycles; strategy-averaging plateaus ≈ 0.56.

HUNL exploitability is intractable (~10^{161} info sets), so the same recipe is probed on Leduc (936 states, exact).



≈8× below random, ≈9× above tuned NFSP — methodology, not magnitude.

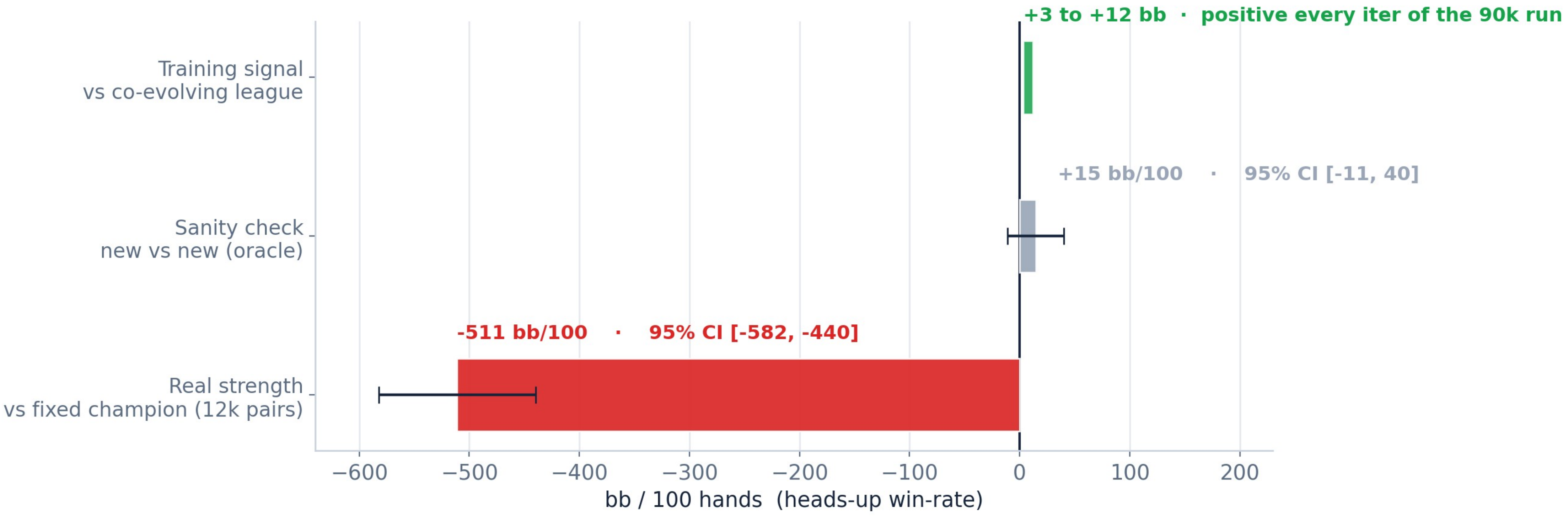


“Convergence” is not low exploitability.

- AlphaHoldem reports loss flattening, Elo plateau, and win-rate (+111.6 mbb/h vs Slumbot). It never computes exploitability.
- Win-rate ≠ exploitability — Lisý & Bowling: two bots tied head-to-head (~20 mbb/g) yet ~1300 mbb/g apart in exploitability.
- The Leduc study is the test HUNL cannot run — not a failed reproduction.

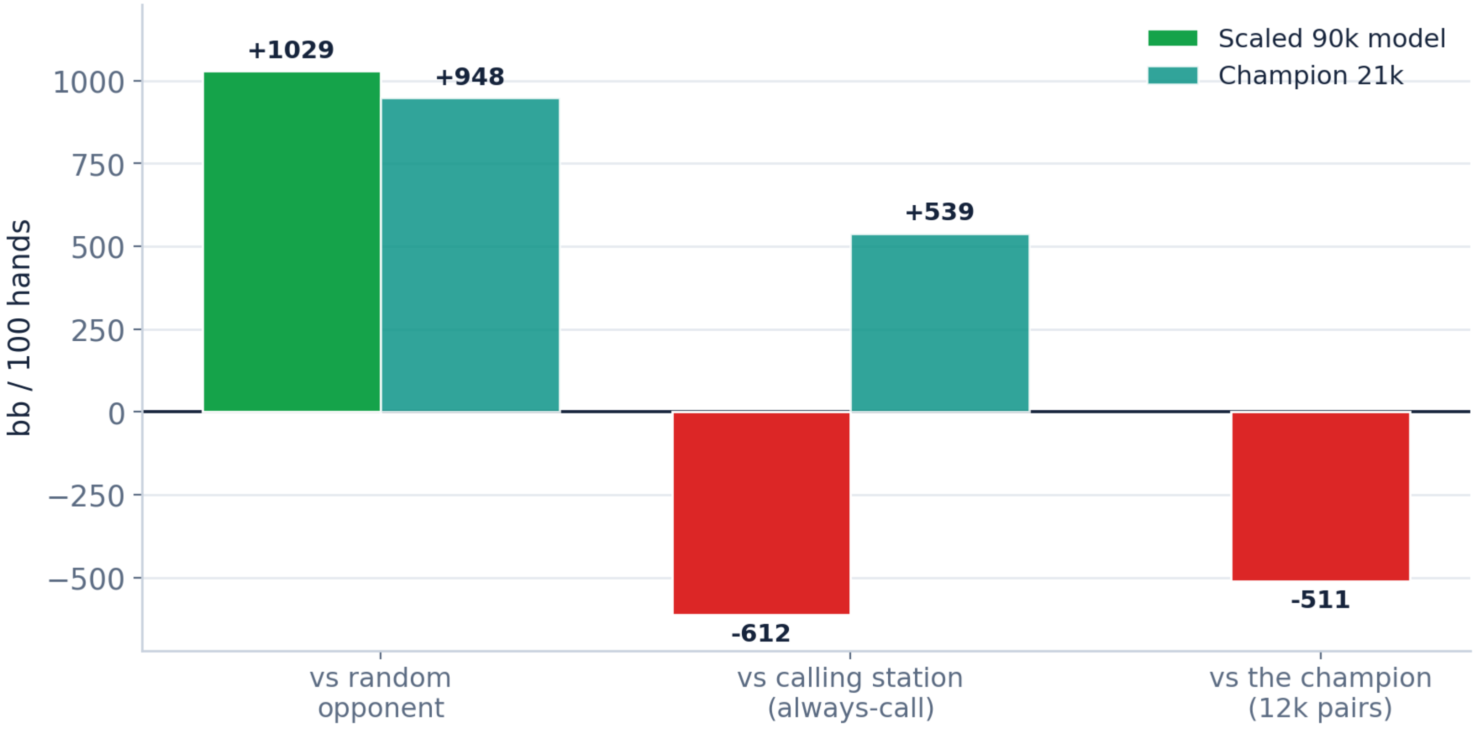
Regressed –511 bb/100 against the champion it started from.

90k GPU iterations (~9×10^7 hands) · vectorized collector (30–100×) · PFSP fix · Trinal-Clip · warm-started from the 21k champion



Over-aggression drift, not collapse.

No calling opponents in the training mix; the co-evolving league learned to fold to aggression.



LIMITATIONS

Strong-but-exploitable, measured carefully.

- Not near-Nash, not a record win-rate.
- Throughput gap vs AlphaHoldem (~2.7B hands, 8 GPUs, 3 days) is not closeable solo.
- The champion remains the strongest model; the scaled run was a negative result.